

ME5412 Robotics for Healthcare

SCI and assessment scales

脊髓损伤 (SCI) 与评估量表.

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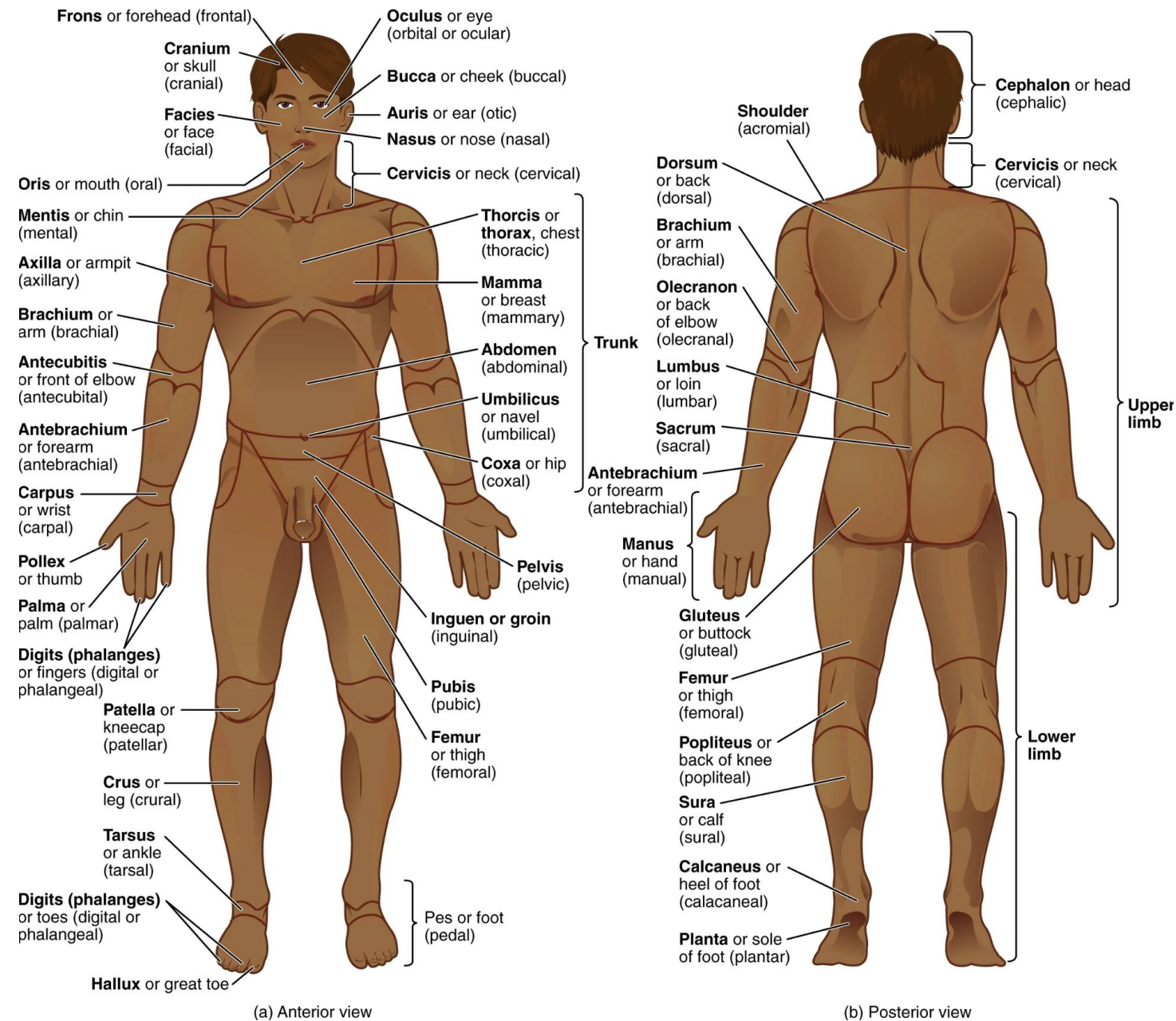
Lecture outline

- Basic concepts and terms of human anatomy
- Spinal cord injuries (SCI)
- Assessment scales

- 人体解剖学基础概念与专业术语
- 脊髓损伤 (SCI)
- 评估量表

Regions of the human body

人体身体分区



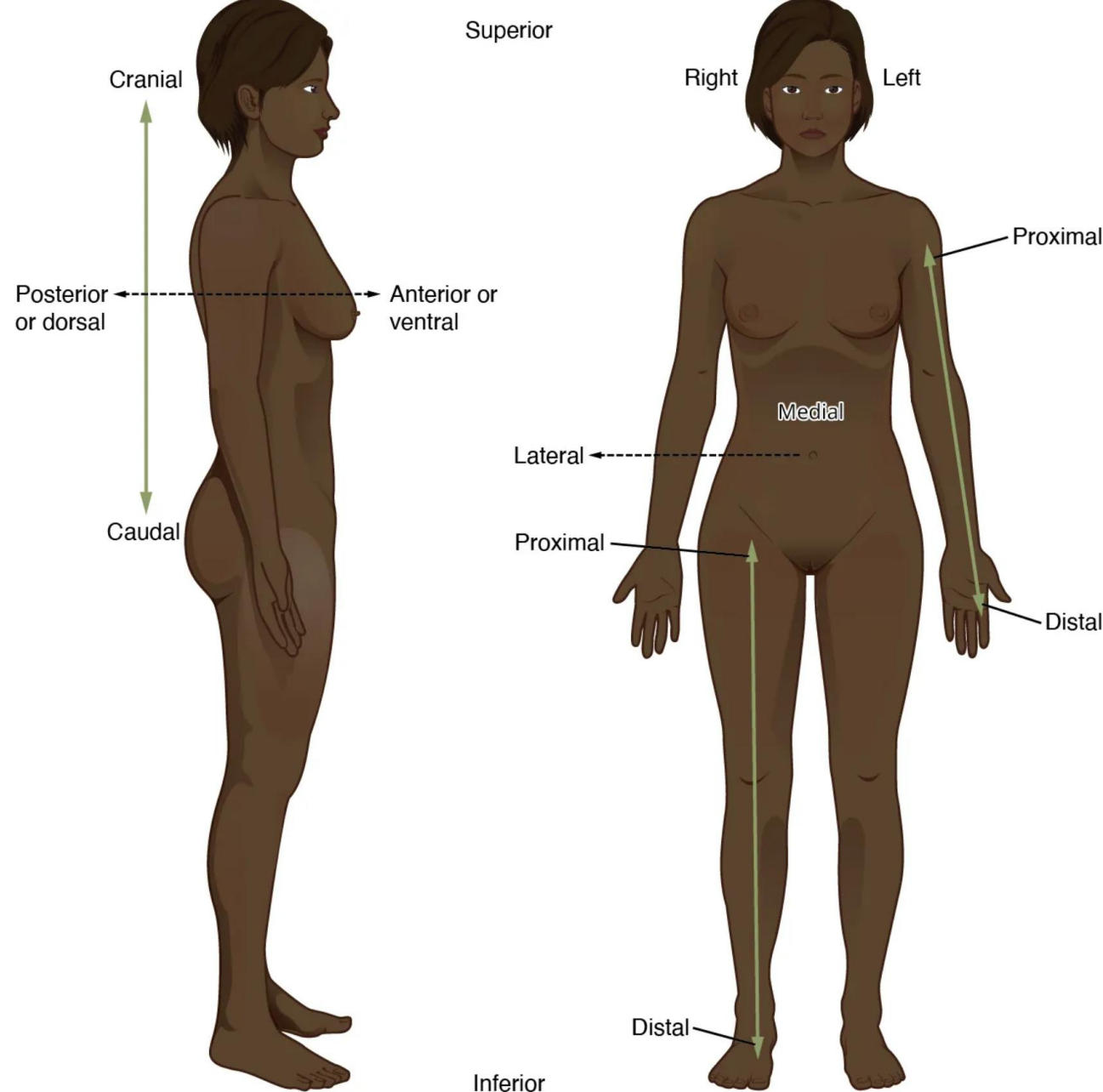
(a) Anterior view

(b) Posterior view

解剖方位术语

Directional Terms

原文	翻译
Anterior (or ventral) Describes the front or direction toward the front of the body. The toes are anterior to the foot.	前侧 (腹侧) 描述朝向身体正面的方位。脚趾在足部的前侧。
Posterior (or dorsal) Describes the back or direction toward the back of the body. The popliteus is posterior to the patella.	后侧 (背侧) 描述朝向身体背部的方位。腘窝位于髌骨 (膝盖骨) 的后方。
Superior (or cranial) describes a position above or higher than another part of the body proper. The orbits are superior to the oris.	上方 (颅侧) 位置高于身体另一结构。眼眶位于嘴巴上方。
Inferior (or caudal) describes a position below or lower than another part of the body proper; near or toward the tail (in humans, the coccyx, or lowest part of the spinal column). The pelvis is inferior to the abdomen.	下方 (尾侧) 位置低于另一身体结构, 靠近尾端 (人体就是尾骨)。骨盆位于腹部下方。
Lateral describes the side or direction toward the side of the body. The thumb (pollex) is lateral to the digits.	外侧 朝向身体侧面。拇指相对于其余手指在外侧。
Medial describes the middle or direction toward the middle of the body. The hallux is the medial toe.	内侧 朝向身体中轴线。大脚趾是足的内侧脚趾。
Proximal describes a position in a limb that is nearer to the point of attachment or the trunk of the body. The brachium is proximal to the antebrachium.	近端 肢体更靠近躯干连接点。上臂相对于前臂是近端。
Distal describes a position in a limb that is farther from the point of attachment or the trunk of the body. The crus is distal to the femur.	远端 肢体远离躯干连接点。小腿相对于大腿属于远端。
Superficial describes a position closer to the surface of the body. The skin is superficial to the bones.	浅表 更靠近身体表面。皮肤在骨骼的浅表侧。
Deep describes a position farther from the surface of the body. The brain is deep to the skull.	深层 远离体表。大脑在颅骨的深层内部。



人体解剖平面

Body Planes

• 矢状面：竖直将身体/器官切分为左右两部分。正中劈开身体中线叫正中矢状面；切出来左右不等大，叫旁矢状面，也叫纵切面。

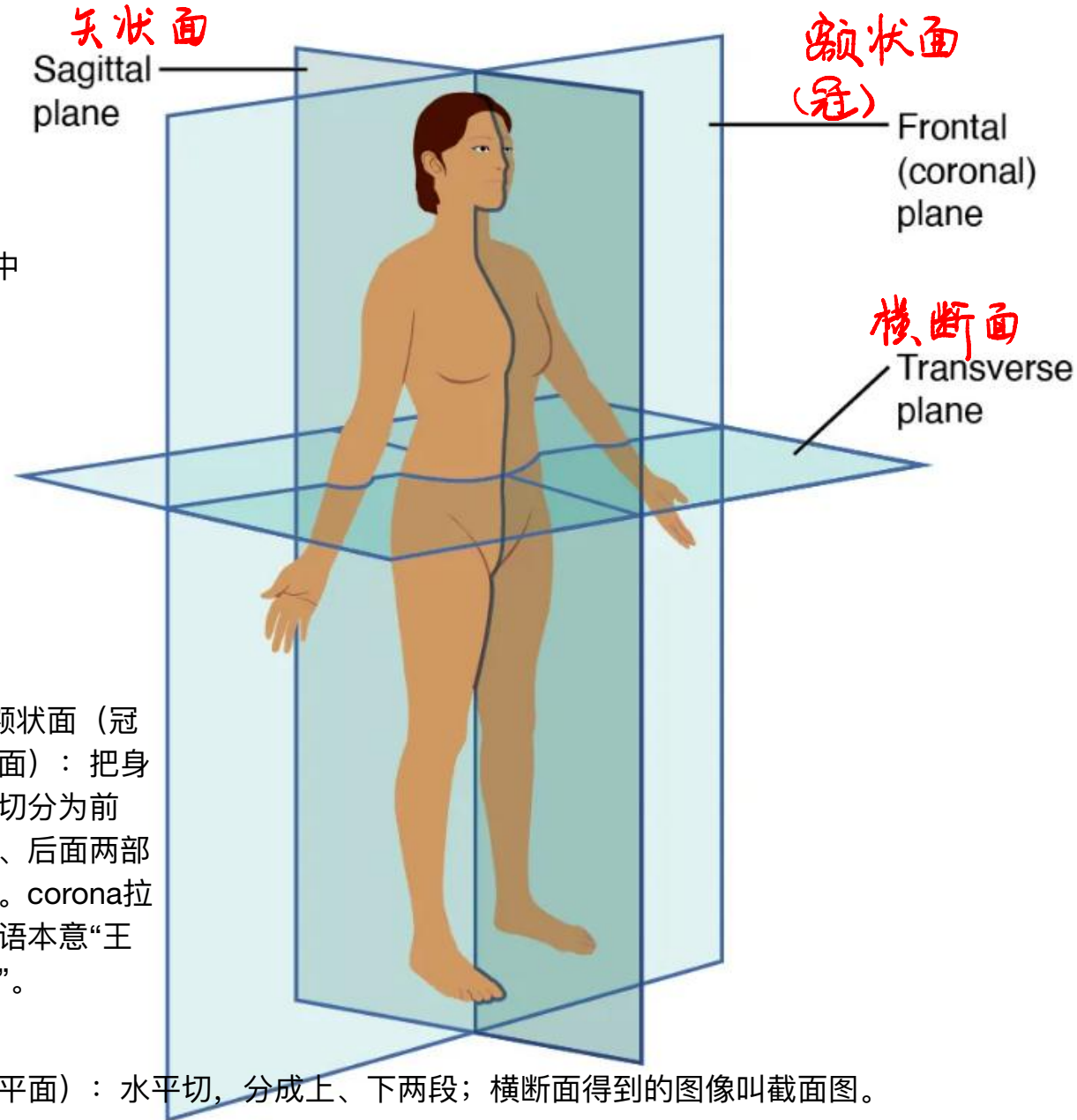
- The **sagittal plane** is the plane that divides the body or an organ vertically into **right and left** sides. If this vertical plane runs directly down the middle of the body, it is called the midsagittal or median plane. If it divides the body into unequal right and left sides, it is called a parasagittal plane or less commonly a longitudinal section.

- The **frontal plane** is the plane that divides the body or an organ into an anterior (**front**) portion and a posterior (**rear**) portion. The frontal plane is often referred to as a **coronal plane**. ("Corona" is Latin for "crown.")

- The **transverse plane** is the plane that divides the body or organ horizontally into **upper and lower** portions. Transverse planes produce images referred to as **cross sections**.

• 额状面（冠状面）：把身体切分为前面、后面两部分。corona拉丁语本意“王冠”。

• 横断面（水平面）：水平切，分成上、下两段；横断面得到的图像叫截面图。

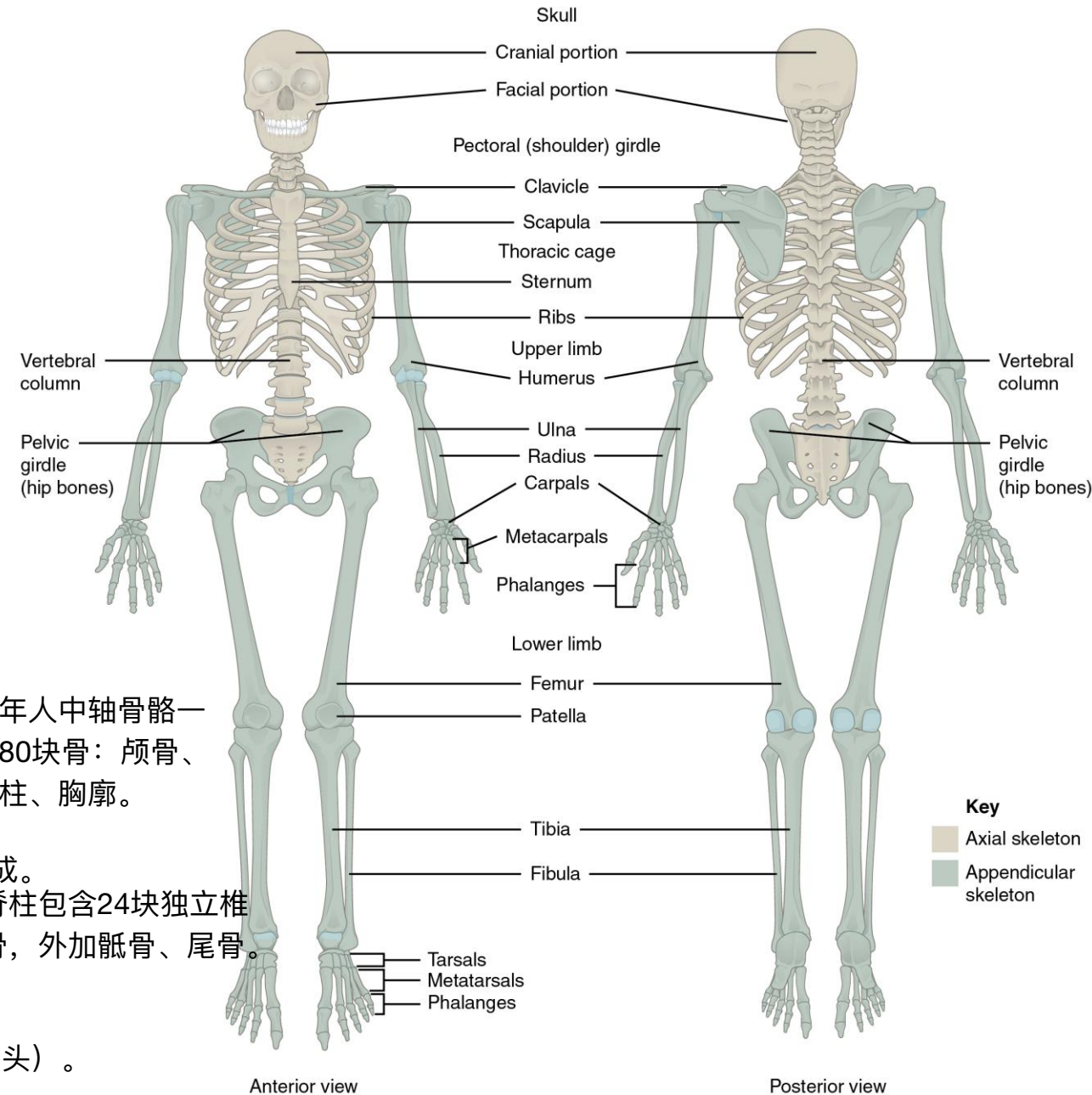


中轴骨骼

Axial skeleton

- 中轴骨骼构成人体竖直中心主轴，包含头、颈、胸、背部全部骨骼。
- 功能：保护大脑、脊髓、心脏、肺；同时作为肌肉附着点，控制头颈背部运动，也连接肩髋带动四肢。
- The **axial skeleton** forms the vertical, central axis of the body and includes all bones of the head, neck, chest, and back.
- It serves to protect the brain, spinal cord, heart, and lungs. It also serves as the attachment site for muscles that move the head, neck, and back, and for muscles that act across the shoulder and hip joints to move their corresponding limbs.
- The axial skeleton of the adult consists of 80 bones, including the **skull**, the **vertebral column**, and the **thoracic cage**.
 - The skull is formed by 22 bones. 颅骨由22块骨头组成。
 - The vertebral column consists of 24 bones, each called a **vertebra**, plus the **sacrum** and **coccyx**. 脊柱包含24块独立椎骨，外加骶骨、尾骨。
 - The thoracic cage includes the 12 pairs of **ribs**, and the **sternum**, the flattened bone of the anterior chest. 胸廓包含12对肋骨 + 胸骨（胸前扁平骨头）。

成年人中轴骨骼—
共80块骨：颅骨、
脊柱、胸廓。



Anterior view

Posterior view

关节

Joints

- 关节按照它能够实现的运动轴数目分为三类；
- 解剖学运动轴对应前面的三个解剖平面（横断面、冠状面、矢状面）。
- 关节分类
 1. 单轴关节：仅允许绕1根轴，在单个平面运动。
肘关节（只能弯曲、伸直）就是单轴关节。
 2. 双轴关节：可以在两个平面运动。
手指掌指关节（指根关节）：可以弯曲伸直，还可以手指分开、并拢，属于双轴关节。
 3. 多轴关节（三轴关节）：可沿三根轴做多方向运动。
肩关节、髋关节。能使上肢或下肢前后摆动、内外摆动，还可以沿着肢体长轴旋转。这种第三种运动使肢体发生旋转，使其前表面朝向或远离身体的正中线。

- Joints are divided into three categories, based on the number of axes of motion provided by each.
- An axis in anatomy is described as the movements in reference to the three anatomical planes: transverse, frontal, and sagittal.
- Joints are classified as:
 - **Uniaxial joints:** they only allow for a motion in a single plane (around a single axis)
 - The elbow joint, which only allows for bending or straightening, is an example of a uniaxial joint
 - **Biaxial joints:** they allow for motions within two planes
 - An example of a biaxial joint is a metacarpophalangeal joint (knuckle joint) of the hand. The joint allows for movement along one axis to produce bending or straightening of the finger, and movement along a second axis, which allows for spreading of the fingers away from each other and bringing them together
 - **Multiaxial joints** (polyaxial or triaxial joints): they allow for the several directions of movement, for movement along three axes
 - The shoulder and hip joints are multiaxial joints. They allow the upper or lower limb to move in an anterior-posterior direction and a medial-lateral direction. In addition, the limb can also be rotated around its long axis. This third movement results in rotation of the limb so that its anterior surface is moved either toward or away from the midline of the body.

具体关节形态分类.

Types of Joints

(a) 枢轴关节：绕一根轴旋转；例第1、2颈椎，实现头部左右转头。

(b) 铰链关节（屈戌关节）：肘关节，如同房门合页，只能屈伸。

(a) **Pivot joints** allow for rotation around an axis, such as between the first and second cervical vertebrae, which allows for side-to-side rotation of the head.

(b) The **hinge joint** of the elbow works like a door hinge.

(c) The articulation between the trapezium carpal bone and the first metacarpal bone at the base of the thumb is a **saddle joint**. (c)鞍状关节：大拇指根部（大多角骨-第一掌骨）。

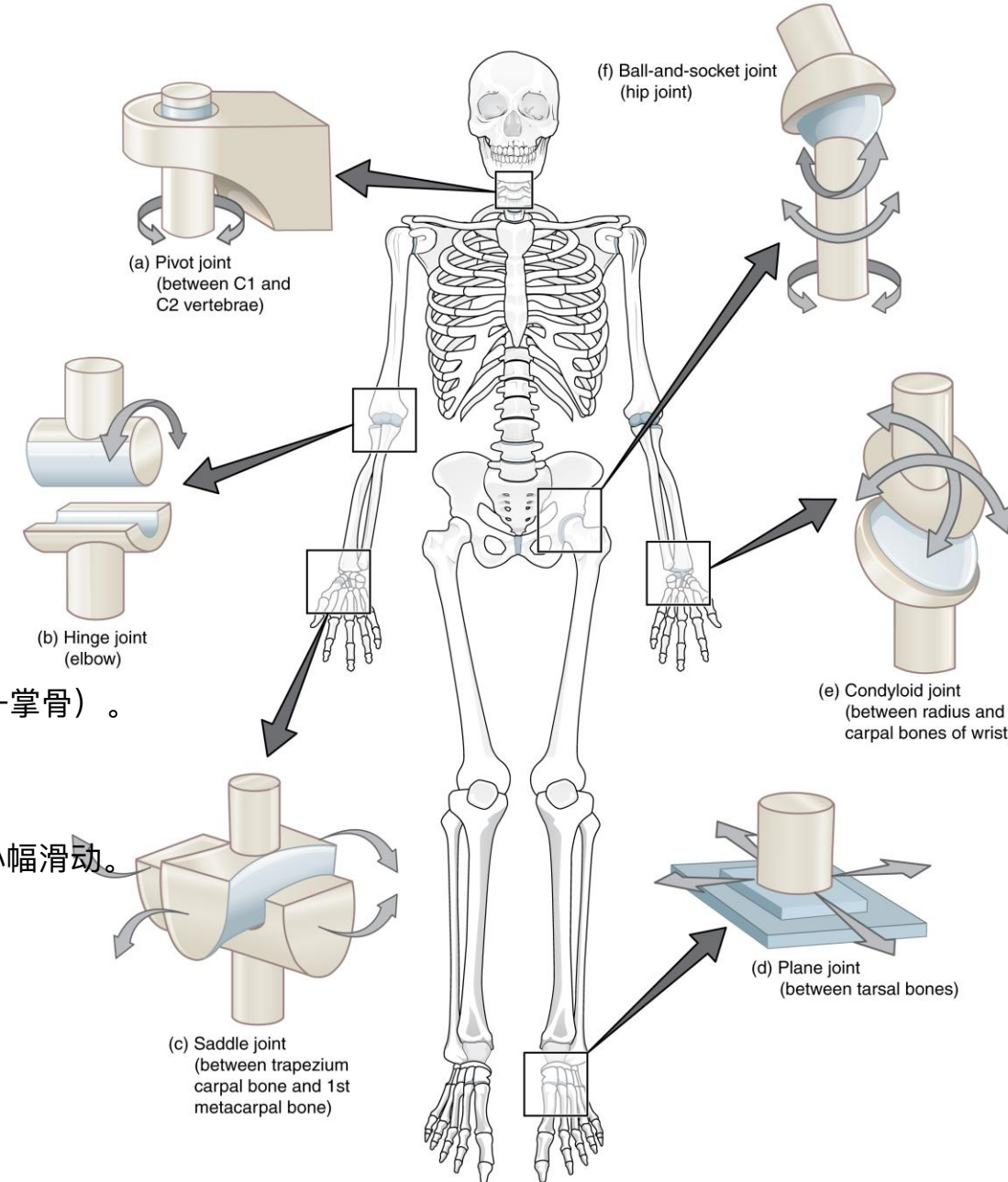
(d) **Plane joints**, such as those between the tarsal bones of the foot, allow for limited gliding movements between bones. (d)平面关节（滑动关节），足跗骨之间；骨头之间做小幅滑动。

(e) The radiocarpal joint of the wrist is a **condyloid joint**.

(f) The hip and shoulder joints are the only **ball-and-socket joints** of the body.

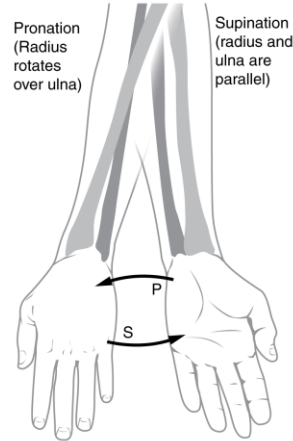
e) 踝状关节（椭圆关节）：手腕桡腕关节。

(f) 球窝关节：人体只有髋关节、肩关节，是活动度最大的关节。



Types of movements

旋前、旋后 (手掌翻转向下、向上)



(g) Pronation (P) and supination (S)

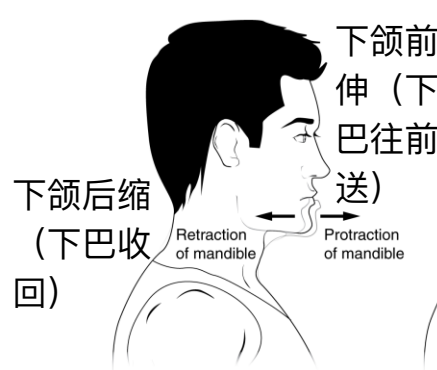
背屈: 脚尖向上勾起

Dorsiflexion

跖屈: 脚尖向下踩 (踮脚动作)

(h) Dorsiflexion and plantar flexion

足内翻、足外翻
(i) Inversion and eversion



(j) 前伸与后缩 (下颌)

(j) Protraction and retraction



(k) 上提与下降 (下颌)

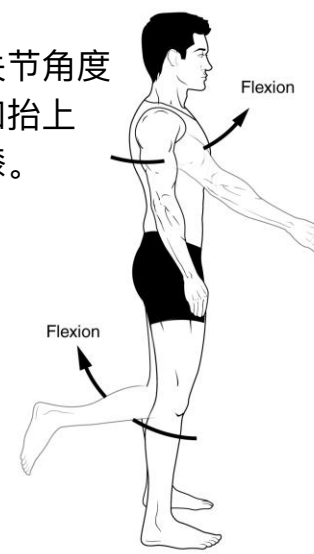
(k) Elevation and depression



(l) Opposition

对掌: 大拇指可以碰到其余指尖, 人手特有。

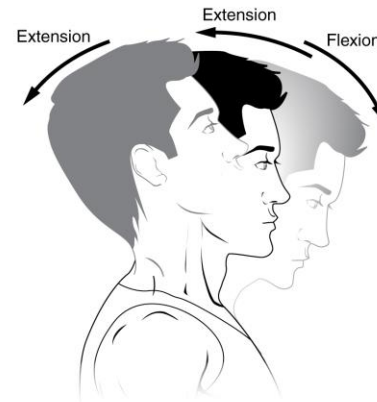
屈曲: 关节角度变小, 如抬上臂、屈膝。



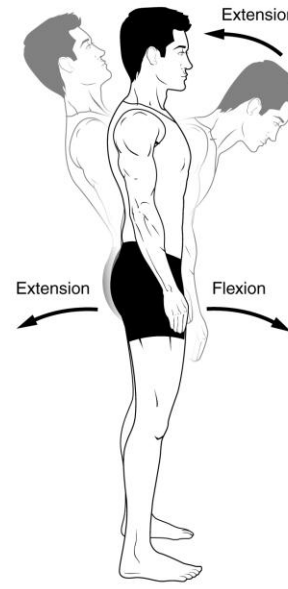
(a) and (b) Angular movements: flexion and extension at the shoulder and knees
(a)(b) 角运动: 肩关节与膝关节处的屈曲、伸展



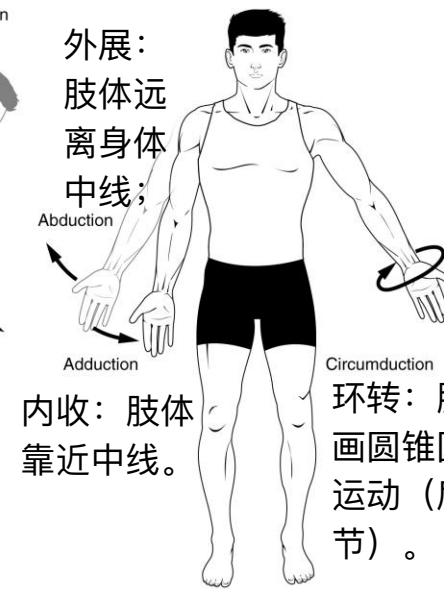
伸展: 关节角度打开, 伸直肢体。



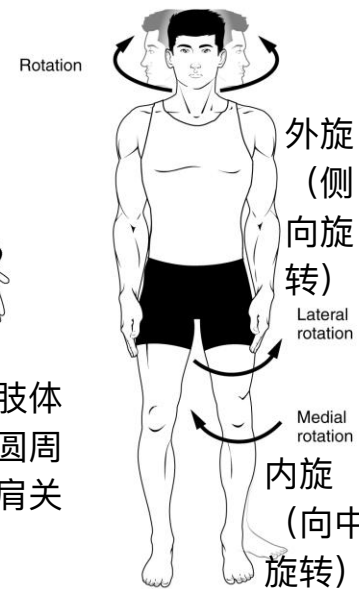
(c) 角运动: 颈部的屈曲、伸展
(c) Angular movements: flexion and extension of the neck



(d) Angular movements: flexion and extension of the vertebral column



(e) Angular movements: abduction, adduction, and circumduction of the upper limb at the shoulder



(f) Rotation of the head, neck, and lower limb
(f) 头、颈部及下肢的旋转

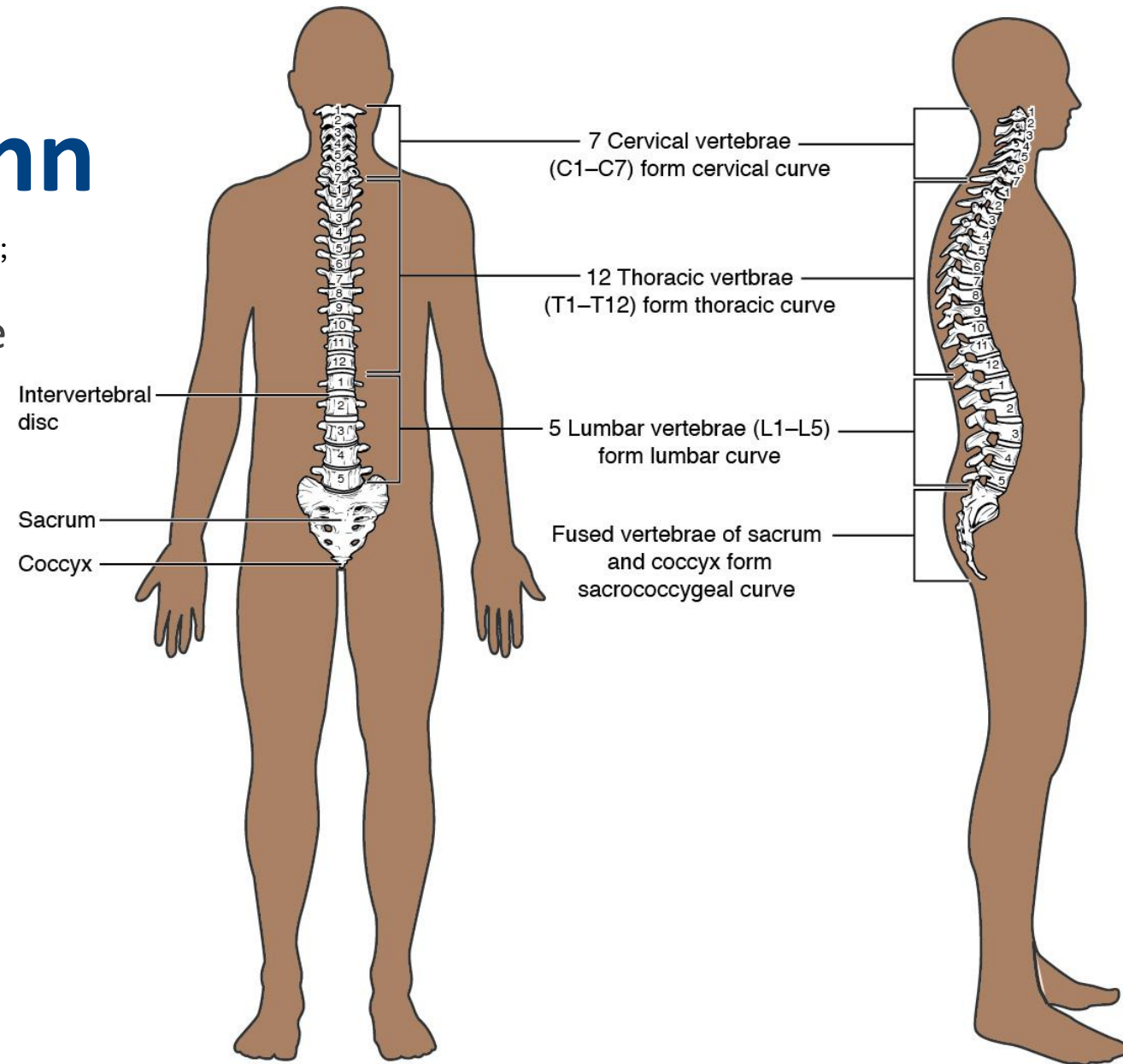
脊柱 (脊椎列)

The vertebral column

脊柱 (spine), 由一串椎骨 (单数vertebra, 复数vertebrae) 组成; 椎骨之间靠椎间盘隔开、连接。椎骨+椎间盘共同构成完整脊柱。

- Also known as the spinal column or spine
- It consists of a sequence of vertebrae (singular = vertebra), each of which is separated and united by an **intervertebral disc**. Together, the vertebrae and intervertebral discs form the vertebral column.
- It is a flexible column that supports the head, neck, and body and allows for their movements.
- It also protects the spinal cord, which passes down the back through openings in the vertebrae.

柔性支撑头部躯干, 允许弯曲运动;
保护从椎骨孔洞中穿过的脊髓。



脊椎分段详解

Regions of the vertebral column

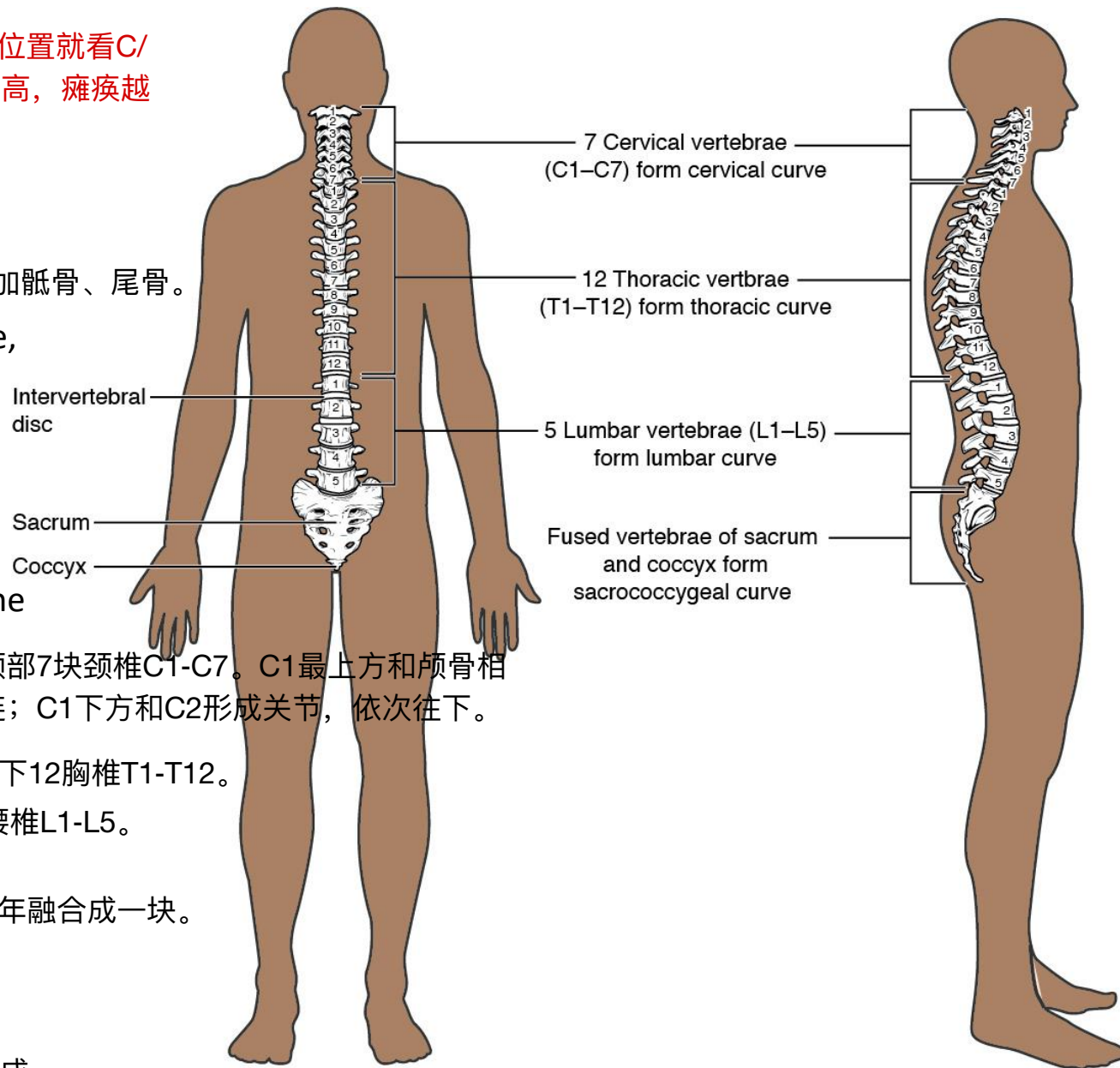
重点：脊髓损伤位置就看C/T/L/S编号，C越高，瘫痪越严重。

胚胎发育早期一共33块椎骨；成年之后，部分融合，剩下24块独立椎骨，外加骶骨、尾骨。

- The vertebral column originally develops as a series of 33 vertebrae, but this number is eventually reduced to 24 vertebrae, plus the sacrum and coccyx.
- The vertebral column is subdivided into **five regions**, with the vertebrae in each area named for that region and numbered in **descending** order. 脊柱分为5大段；每段椎骨编号从上往下数。
- In the neck, there are **7 cervical vertebrae**, each designated with the letter “C” followed by its number. Superiorly, the C1 vertebra articulates (forms a joint) with the occipital condyles of the skull. Inferiorly, C1 articulates with the C2 vertebra, and so on.
- Below these are the **12 thoracic vertebrae**, designated **T1–T12**. 往下12胸椎T1-T12。
- The lower back contains the **L1–L5 lumbar vertebrae**. 下背部5块腰椎L1-L5。
- The single **sacrum**, which is also part of the pelvis, is formed by the **fusion of five sacral vertebrae**. 骶骨属于骨盆一部分，胚胎5块骶椎成年融合成一块。
- The **coccyx**, or tailbone, results from the **fusion of four small coccygeal vertebrae**. However, the sacral and coccygeal fusions do not start until age 20 and are not completed until middle age.

尾骨，4块尾椎融合而成；骶骨、尾骨的融合从20岁才开始，中年才完全完成。

颈部7块颈椎C1-C7。C1最上方和颅骨相连；C1下方和C2形成关节，依次往下。



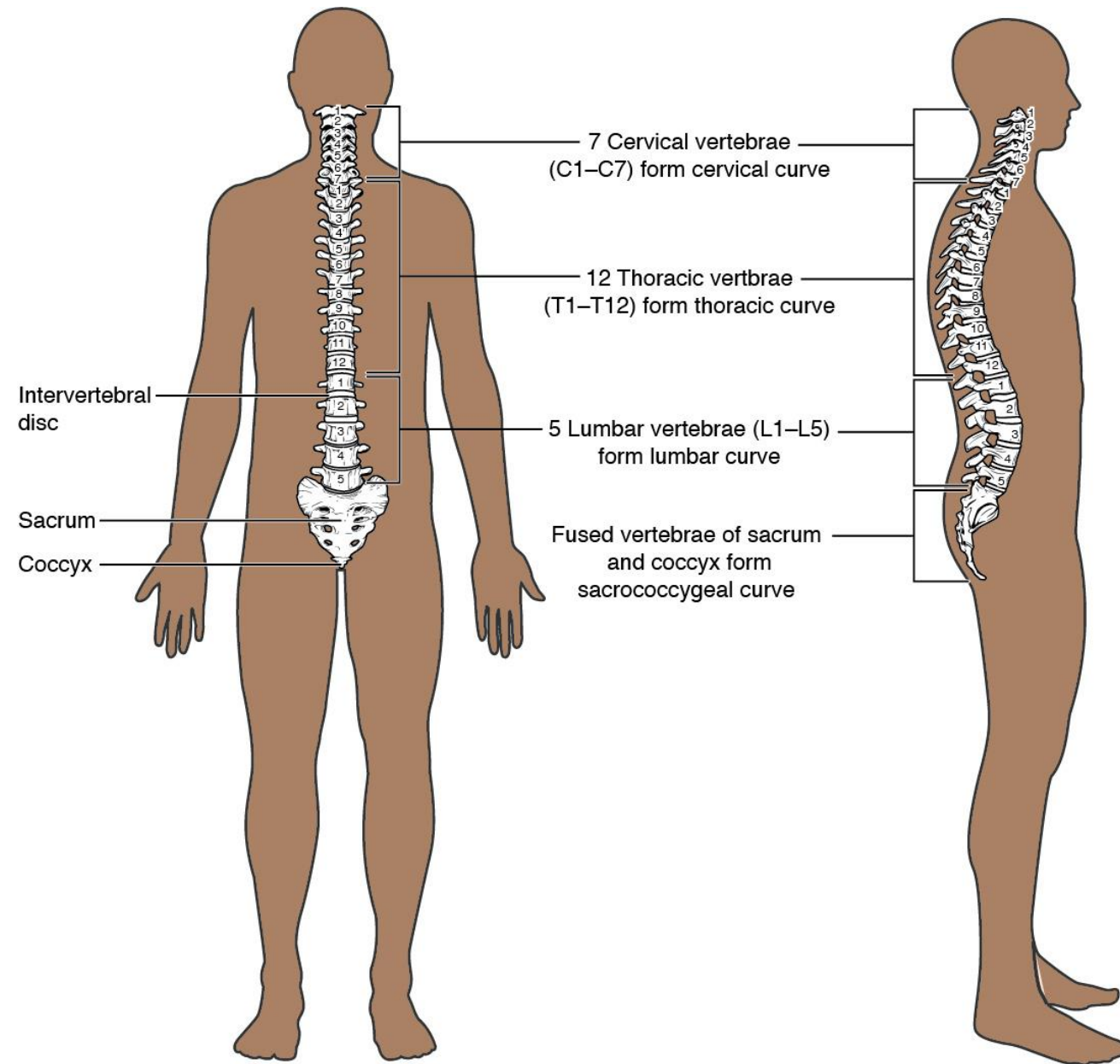
An interesting anatomical fact is that almost all mammals have seven cervical vertebrae, regardless of body size. This means that there are large variations in the size of cervical vertebrae, ranging from the very small cervical vertebrae of a shrew to the greatly elongated vertebrae in the neck of a giraffe. In a full-grown giraffe, each cervical vertebra is 11 inches

脊椎的生理弯曲.

Curvatures of the vertebral column

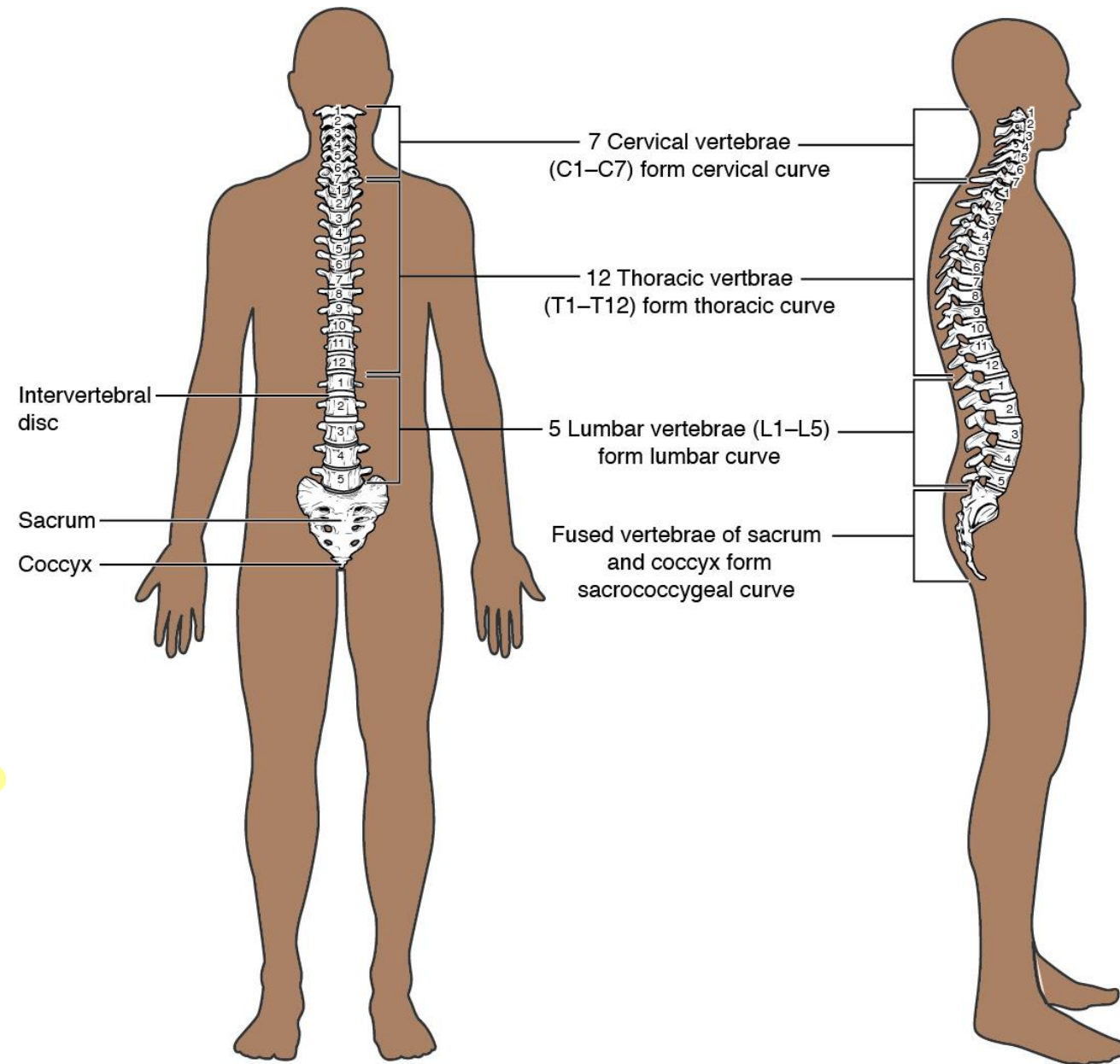
成年人脊柱不是直线，一共存在四处生理弯曲。

- The adult vertebral column does not form a straight line, but instead has **four curvatures** along its length.
- These curves increase the vertebral column's strength, flexibility, and ability to absorb shock. When the load on the spine is increased, by carrying a heavy backpack for example, the curvatures increase in depth (become more curved) to accommodate the extra weight. They then spring back when the weight is removed.
- The four adult curvatures are classified as either **primary** or **secondary** curvatures. Primary curves are retained from the original fetal curvature, while secondary curves develop after birth.
- In the adult, fetal curvature is retained in two regions of the vertebral column as the **thoracic curve**, which involves the thoracic vertebrae, and the **sacroccocygeal curve**, formed by the sacrum and coccyx.
- Secondary curves are concave posteriorly, opposite in direction to the original fetal curvature. The **cervical curve** of the neck region develops as the infant begins to hold their head upright when sitting. Later, as the child begins to stand and then to walk, the **lumbar curve** of the lower back develops. In adults, the lumbar curve is generally deeper in females.



Curvatures of the vertebral column

- 成人脊柱并非呈直线，而是沿其长度方向具有四个生理弯曲。
- 这些弯曲增强了脊柱的强度、灵活性以及减震能力。当脊柱承受的负荷增加时（例如背负沉重的背包），这些弯曲会加深（变得更加弯曲），以适应额外的重量。当重量移除后，会恢复原状。
- 成人脊柱的四种弯曲可分为原发性弯曲和继发性弯曲。原发性弯曲保留了胎儿期原有的弯曲形态，而继发性弯曲则是在出生后逐渐形成的。
- 在成人中，胎儿期的弯曲在脊柱的两个区域得以保留：一个是胸椎部的胸曲，涉及胸椎；另一个是骶尾部的骶尾曲，由骶骨和尾骨构成。
- 继发性弯曲向后凹陷，方向与原始胎儿弯曲相反。颈部的颈椎曲在婴儿开始坐立并能直立头部时形成。随后，随着儿童开始站立并行走，下背部的腰曲逐渐发育。在成年人中，女性的腰曲通常比男性更深。

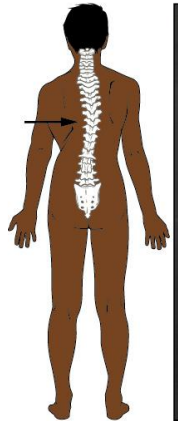
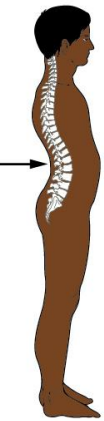


脊柱异常弯曲 (脊柱病变)

Disorders of the vertebral column

发育异常、病理改变或者肥胖，会让脊柱原本的弯曲加剧，造成异常、过度的脊柱弯曲。

- **后凸畸形 (驼背)**：胸椎向后弯曲程度过大。骨质疏松造成上位胸椎前部骨质削弱、被侵蚀，椎骨慢慢塌陷，会引发该病症。
- **前凸畸形 (鞍背)**：腰椎向前弯曲过度；常见于肥胖或者妊娠晚期。腹部重量堆积，身体重力线向前偏移，造成骨盆前倾，腰曲显著加深。
Developmental anomalies, pathological changes, or obesity can enhance the normal vertebral column curves, resulting in the development of abnormal or excessive curvatures.
- **Kyphosis**, also referred to as humpback or hunchback, is an **excessive posterior curvature of the thoracic region**. This can develop when osteoporosis causes weakening and erosion of the anterior portions of the upper thoracic vertebrae, resulting in their gradual collapse.
- **Lordosis**, or swayback, is an **excessive anterior curvature of the lumbar region** and is most commonly associated with obesity or late pregnancy. The accumulation of body weight in the abdominal region results in an anterior shift in the line of gravity that carries the weight of the body. This causes an anterior tilt of the pelvis and a pronounced enhancement of the lumbar curve.
- **Scoliosis** is an abnormal, **lateral curvature**, accompanied by twisting of the vertebral column. Compensatory curves may also develop in other areas of the vertebral column to help maintain the head positioned over the feet. Scoliosis is the most common vertebral abnormality among girls. The cause is usually unknown, but it may result from weakness of the back muscles, defects such as differential growth rates in the right and left sides of the vertebral column, or differences in the length of the lower limbs. When present, scoliosis tends to get worse during adolescent growth spurts. Although most individuals do not require treatment, a back brace may be recommended for growing children. In extreme cases, surgery may be required.



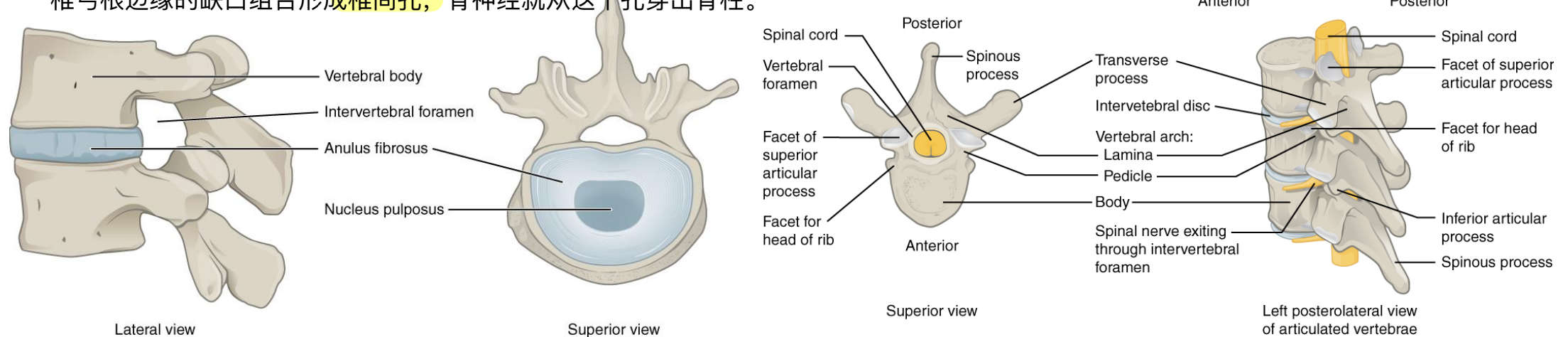
- **脊柱侧凸**：脊柱异常向侧方弯曲，同时伴随椎体旋转。脊柱其他节段会出现代偿弯曲，保证头部能够对齐足部。脊柱侧凸是女孩最高发的脊柱畸形。多数病因不明；诱因包括背部肌肉力量薄弱、脊柱左右两侧生长速度不一致、双下肢长度不等。青春期快速发育阶段，侧弯往往加重。多数患者无需治疗；还在发育的儿童建议佩戴支具；严重病例需要手术。

单块椎骨的结构

Structure of a Vertebra

- 脊柱不同节段，椎骨大小形态不一样，但是基础结构模式一致。一块典型椎骨包含椎体、椎弓、七个突起。
- 椎体是椎骨靠前的部分，承担身体重量。顺着脊柱往下，椎体体积、厚度逐步增大。相邻椎体之间靠椎间盘隔开，同时牢固连接。

- Within the different regions of the vertebral column, vertebrae vary in size and shape, but they all follow a similar structural pattern. A typical vertebra will consist of a **body**, a **vertebral arch**, and **seven processes**.
- The **body** is the anterior portion of each vertebra and is the part that supports the body weight. Because of this, the vertebral bodies progressively increase in size and thickness going down the vertebral column. The bodies of adjacent vertebrae are separated and strongly united by an **intervertebral disc**.
- The large opening between the vertebral arch and body is the **vertebral foramen**, which contains the **spinal cord**. In the intact vertebral column, the vertebral foramina of all of the vertebrae align to form the **vertebral (spinal) canal**, which serves as the bony protection and passageway for the spinal cord down the back. When the vertebrae are aligned together in the vertebral column, notches in the margins of the pedicles of adjacent vertebrae together form an **intervertebral foramen**, the opening through which a spinal nerve exits from the vertebral column.
- 椎弓和椎体之间的大孔洞叫做椎孔，容纳脊髓。完整脊柱，全部椎孔对齐连成椎管，骨性管道保护脊髓，供脊髓向下通行。椎骨叠在一起，相邻椎弓根边缘的缺口组合形成椎间孔，脊神经就从这个孔穿出脊柱。



椎间盘

- 椎间盘是纤维软骨垫，填充相邻椎体之间空隙。
- 每一块椎间盘固定在上下椎体上，把椎骨牢固连接。
- 负重的时候椎间盘起到缓冲垫的作用。颈椎的椎间盘比较薄；承担最大体重的腰椎，椎间盘厚度最大。

Intervertebral discs

- 髓核：质地更软、胶冻样物质，含水量很高，可以抵抗压力，对负重至关重要。
- 髓核胶冻特性，允许椎间盘变形；脊柱运动，椎骨相对相邻椎骨前后、左右摆动。身体向前弯腰时，椎间盘前侧受压，后侧会扩张。

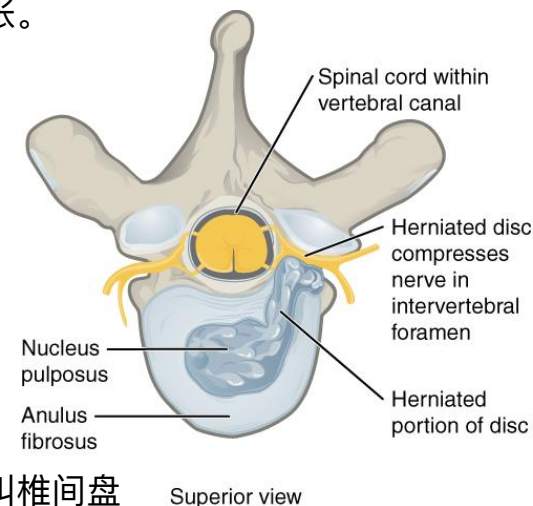
- An **intervertebral disc** is a fibrocartilaginous pad that fills the gap between adjacent vertebral bodies.
- Each disc is anchored to the bodies of its adjacent vertebrae, thus strongly uniting these.
- The discs also provide padding between vertebrae during weight bearing. Because of this, intervertebral discs are thin in the cervical region and thickest in the lumbar region, which carries the most body weight.
- The **nucleus pulposus** consists of a softer, more gel-like material. It has a high water content that serves to resist compression and thus is important for weight bearing.

- The gel-like nature of the nucleus pulposus also allows the intervertebral disc **to change shape** as one vertebra rocks side to side or forward and back in relation to its neighbors during movements of the vertebral column.

Thus, bending forward causes compression of the anterior portion of the disc but expansion of the posterior

disc. • 如果后方纤维环因为损伤、年纪增长强度下降；弯腰搬重物，椎间盘受压，髓核向后突破纤维环，造成椎间盘突出（也叫椎间盘破裂、滑脱）。向后鼓出的髓核会压迫从椎间孔穿出的脊神经；该神经支配区域会出现疼痛、肌肉无力。

- If the posterior anulus fibrosus is weakened due to injury or increasing age, the pressure exerted on the disc when bending forward and lifting a heavy object can cause the nucleus pulposus to protrude posteriorly through the anulus fibrosus, resulting in a **herniated disc** ("ruptured" or "slipped" disc). The posterior bulging of the nucleus pulposus can cause **compression of a spinal nerve** at the point where it exits through the intervertebral foramen, with resulting pain and/or muscle weakness in those body regions supplied by that nerve.
- The most common sites for disc herniation are the L4/L5 or L5/S1 intervertebral discs, which can cause sciatica, a widespread pain that radiates from the lower back down the thigh and into the leg. Similar injuries of the C5/C6 or C6/C7 intervertebral discs, following forcible hyperflexion of the neck from a collision accident or football injury, can produce pain in the neck, shoulder, and upper limb.



- 椎间盘突出最高发位置L4/L5、L5/S1，会引发坐骨神经痛：疼痛从下背部放射到大腿、小腿。车祸、球类撞击造成颈部猛烈过度前屈，C5/C6、C6/C7椎间盘受损，会引发颈、肩、上肢疼痛。

脊髓损伤

Spinal cord injuries (SCI)

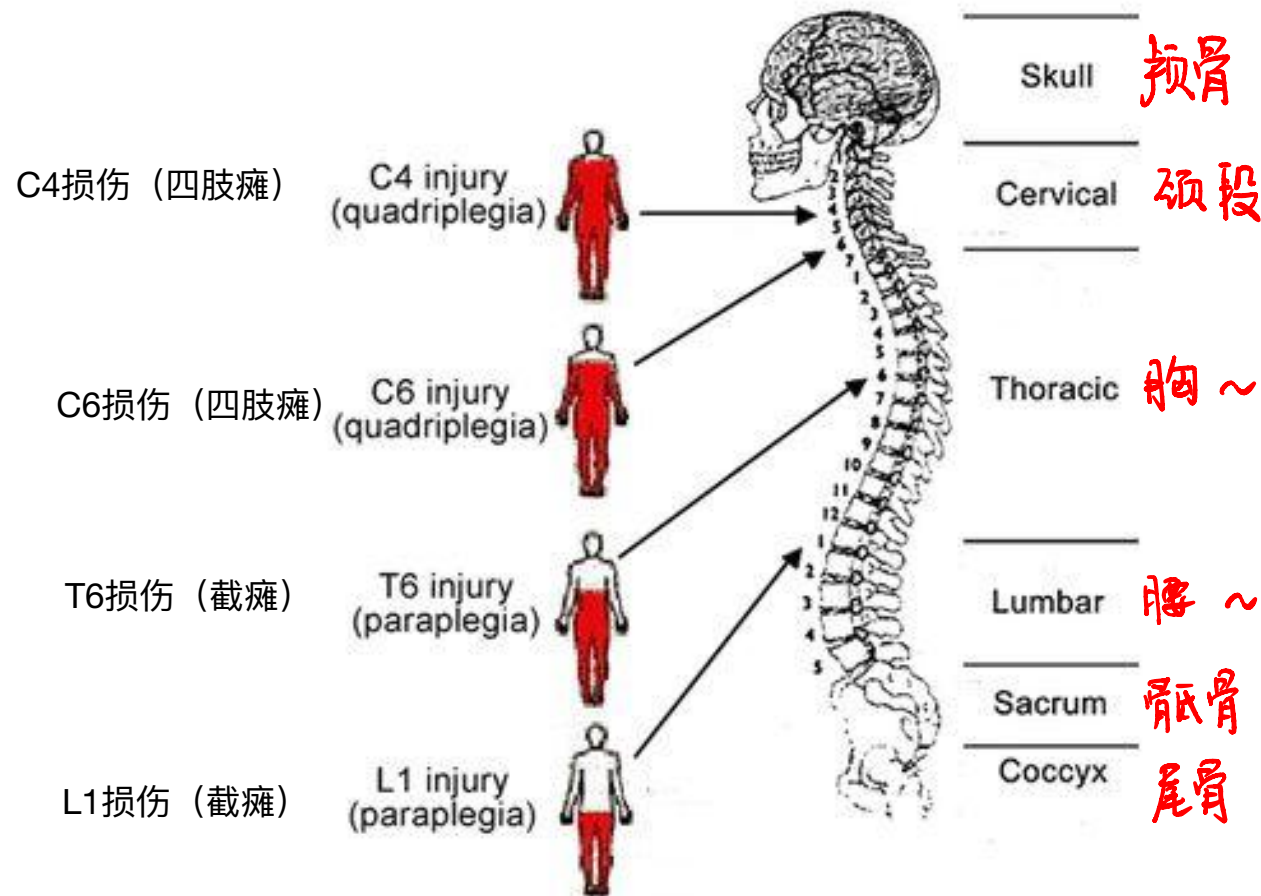


脊髓损伤症状:

SCI symptoms

损伤平面与瘫痪程度

Level of Injury and Extent of Paralysis



VERTEBRAL LEVEL	NERVE ROOT*	INNERVATION	POSSIBLE SYMPTOMS
C1	C1	Intracranial Blood Vessels	Headaches + Migraine Headaches
C2	C2	• Eyes • Lacrimal Gland	• Dizziness • Sinus Problems
C3	C3	• Parotid Gland • Scalp	• Allergies • Head Colds • Fatigue
C4	C4	• Base of Skull • Neck Muscles • Diaphragm	• Vision Problems • Runny Nose
C5	C5	• Neck Muscles • Shoulders	• Sore Throat • Stiff Neck
C6	C6	• Elbows • Arms • Wrists	• Cough • Croup • Arm Pain
C7	C7	• Hands • Fingers • Esophagus • Heart • Lungs • Chest	• Hand and Finger Numbness or Tingling • Asthma • Heart Conditions • High Blood Pressure
C8	C8		
T1	T1	Arms • Esophagus	Wrist, Hand and Finger
T2	T2	• Heart • Lungs • Chest	Numbness or Pain • Middle Back
T3	T3	• Larynx • Trachea	Pain • Congestion • Difficulty Breathing • Asthma • High Blood
T4	T4		Pressure • Heart Conditions
T5	T5	Gallbladder • Liver	• Bronchitis • Pneumonia
T6	T6	• Diaphragm • Stomach	• Gallbladder Conditions
T7	T7	• Pancreas • Spleen	• Jaundice • Liver Conditions
T8	T8	• Kidneys • Small Intestine	• Stomach Problems • Ulcers
T9	T9	• Appendix • Adrenals	• Gastritis • Kidney Problems
T10	T10	Small Intestines • Colon • Uterus	
T11	T11		
T12	T12	Uterus • Colon • Buttocks	
L1	L1	Large Intestines	Constipation • Colitis • Diarrhea
L2	L2	• Buttocks • Groin	• Gas Pain • Irritable Bowel
L3	L3	• Reproductive Organs	• Bladder Problems • Menstrual Problems • Low Back Pain
L4	L4	• Colon • Thighs • Knees	• Pain or Numbness in Legs
L5	L5	• Legs • Feet	
SACRAL	SACRAL	Buttocks • Reproductive Organs • Bladder • Prostate Gland • Legs • Ankles • Feet • Toes	Constipation • Diarrhea • Bladder Problems • Menstrual Problems • Lower Back Pain • Pain or Numbness in Legs

SCI symptoms

SCIs may lead to 4 different clinical conditions:

- **Paraplegia:** sensory-motor syndrome with paralysis of lower limbs, due to SCI of **thoracic or lumbar** spinal cord tract
- **Tetraplegia:** more severe sensory-motor syndrome that adds to paraplegia impairs a paralysis of upper limbs and neurovegetative disorders. It is due to SCI of the **cervical** spinal cord tract.
- **Paraparesis:** incomplete paralysis of lower limbs
- **Tetraparesis:** incomplete paralysis of all limbs

	Complete	Incomplete
Lower limbs	Paraplegia	Paraparesis
Upper and lower limbs	Tetraplegia	Tetraparesis



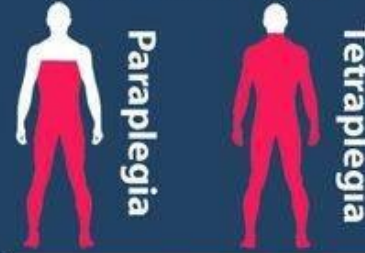
Spinal Cord Injury Infographic Types of Spinal Cord Injury

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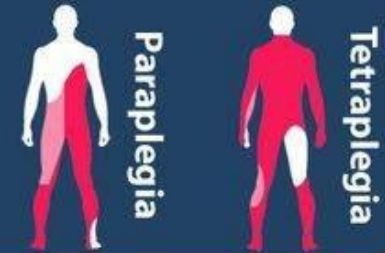
Complete Spinal Cord Injury

Complete loss of motor and sensory function below the spinal cord injury.



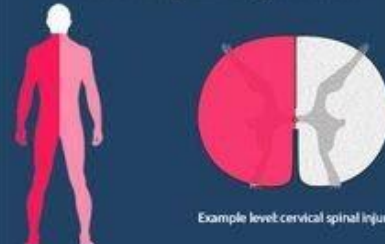
Incomplete Spinal Cord Injury

Partial random preservation of motor or sensory function below the spinal cord injury.



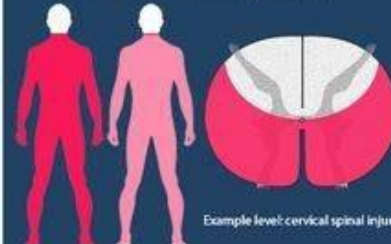
Common Types of Incomplete Spinal Cord Injuries

Brown-Séquard Syndrome



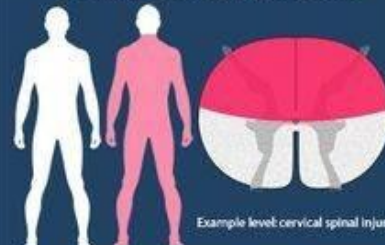
Below injury level, motor weakness or paralysis on one side of the body (hemiparaplegia). Loss of sensation on the opposite side (hemi-anesthesia).

Anterior Cord Syndrome



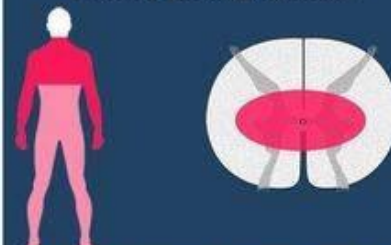
Below injury level, motor paralysis and loss of pain and temperature sensation. Proprioception (position sense), touch and vibratory sensation preserved.

Posterior Cord Syndrome



Below injury level, motor function preserved. Loss of sensory function: pressure, stretch, and proprioception (position sense).

Central Cord Syndrome



Results from cervical spinal injuries. Greater motor impairment in upper body compared to lower body. Variable sensory loss below the level of injury.

Cervical Nerves:

Diaphragm
Deltoids
Biceps
Wrist extensors
Rotates arm
Triceps
Bends fingers

C1
C2
C3
C4
C5
C6
C7

Thoracic Nerves:

Spread fingers
Chest muscles
Abdominal muscles
Muscles in the back

T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12

Lumbar Nerves:

Hip muscles
Thigh muscles
Knee Muscles
Foot muscles

L1
L2
L3
L4
L5

Sacral Nerves:

Bladder and bowel
Sexual function

S1
S2
S3
S4
S5

Cervical

Thoracic

Lumbar

Sacral

Sources: www.apparelyzed.com
www.wikipedia.org
sci.rutgers.edu

Key:

■ Normal Function
■ Impaired Motor Function
■ Impaired Sensory Function

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SCI symptoms

脊髓损伤会造成四种不同临床状态：

- 截瘫：感觉-运动综合征，双下肢瘫痪；由胸段、腰段脊髓损伤造成。
- 四肢瘫：更加严重的感觉运动综合征，除截瘫表现之外，还伴随双上肢瘫痪、自主神经功能紊乱；病因是颈段脊髓损伤。
- 轻截瘫：双下肢不完全瘫痪
- 四肢轻瘫：四肢全部不完全瘫痪

	Complete	Incomplete
Lower limbs	Paraplegia 截瘫	Paraparesis 轻截瘫
Upper and lower limbs	Tetraplegia 四肢瘫	Tetraparesis 四肢轻瘫



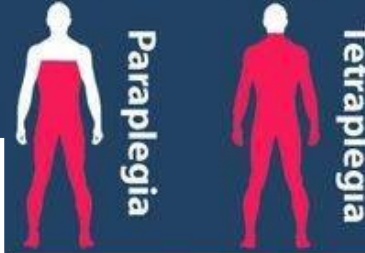
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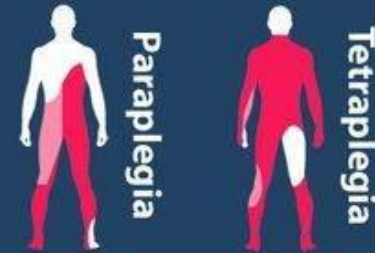
Complete Spinal Cord Injury

Complete loss of motor and sensory function below the spinal cord injury.



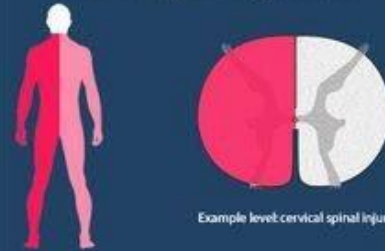
Incomplete Spinal Cord Injury

Partial random preservation of motor or sensory function below the spinal cord injury.

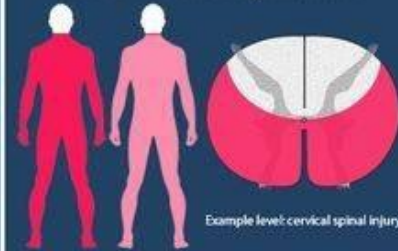


Common Types of Incomplete Spinal Cord Injuries

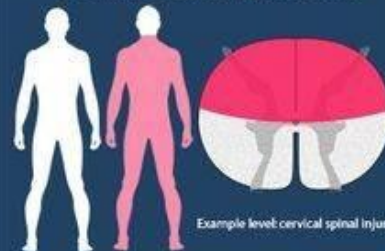
Brown-Séquard Syndrome



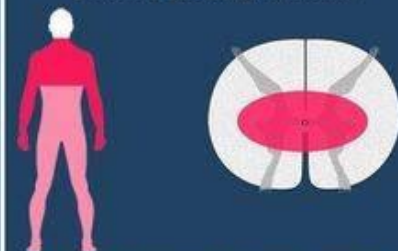
Anterior Cord Syndrome



Posterior Cord Syndrome



Central Cord Syndrome



Cervical Nerves:

Diaphragm
Deltoids
Biceps
Wrist extensors
Rotates arm
Triceps
Bends fingers

C1
C2
C3
C4
C5
C6
C7

Cervical

Thoracic Nerves:

Spread fingers
Chest muscles
Abdominal muscles
Muscles in the back

T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12

Thoracic

Lumbar Nerves:

Hip muscles
Thigh muscles
Knee Muscles
Foot muscles

L1
L2
L3
L4
L5

Lumbar

Sacral Nerves:

Bladder and bowel
Sexual function

S1
S2
S3
S4
S5

Sacral

Sources: www.apparelyzed.com
www.wikipedia.org
sci.rutgers.edu

Key:

■ Normal Function
■ Impaired Motor Function
■ Impaired Sensory Function

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皮节与肌节

Dermatomes and myotomes

- **Dermatomes:**

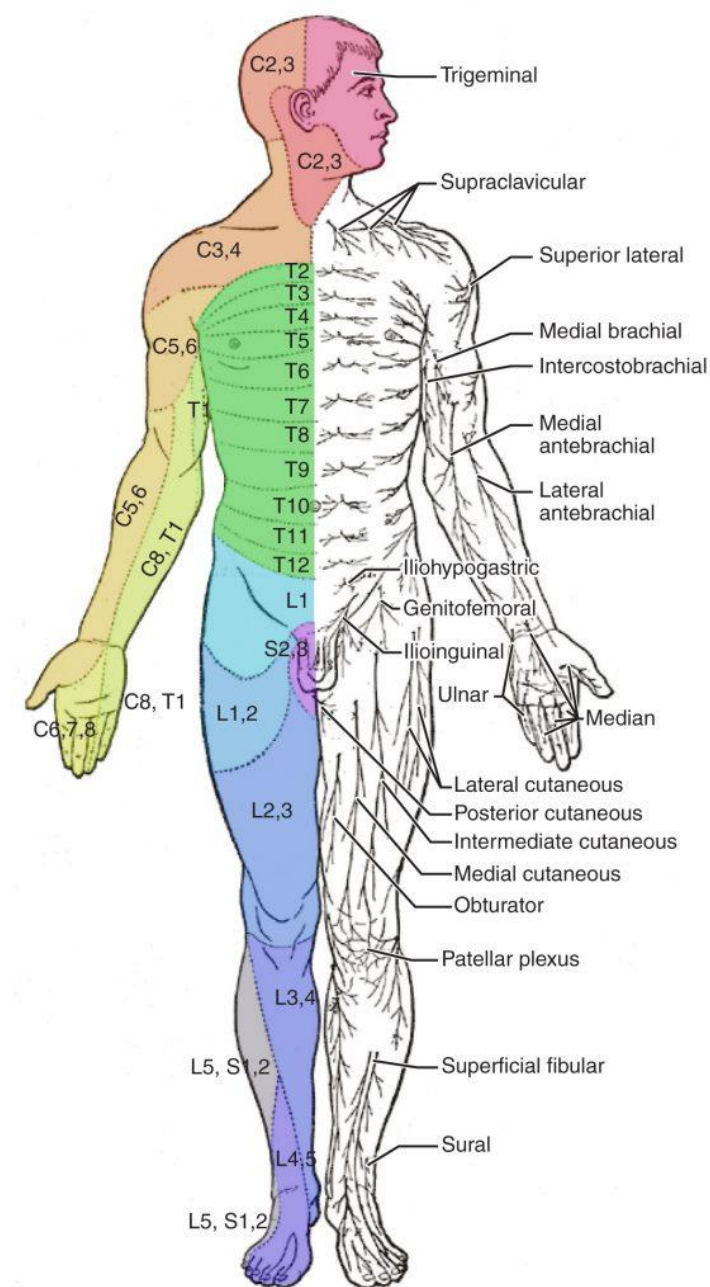
skin area innervated by the **dorsal (sensory)** root of a single spinal nerve

- 皮节：单条脊神经后根（感觉根）所支配的皮肤区域

- **Myotomes:**

muscle ~~skin~~ area innervated by the **ventral (motor)** root of a single spinal nerve

- 肌节：单条脊神经前根（运动根）支配对应的肌肉区域



ASIA scale

STANDARD NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY

MOTOR

KEY MUSCLES

	R	L
C2		
C3		
C4		
C5		
C6		
C7		
C8		
T1		
T2		
T3		
T4		
T5		
T6		
T7		
T8		
T9		
T10		
T11		
T12		
L1		
L2		
L3		
L4		
L5		
S1		
S2		
S3		
S4-5		

Elbow flexors
Wrist extensors
Elbow extensors
Finger flexors (distal phalanx of middle finger)
Finger abductors (little finger)

0 = total paralysis
1 = palpable or visible contraction
2 = active movement, gravity eliminated
3 = active movement, against gravity
4 = active movement, against some resistance
5 = active movement, against full resistance
NT = not testable

Hip flexors
Knee extensors
Ankle dorsiflexors
Long toe extensors
Ankle plantar flexors

☐ Voluntary anal contraction (Yes/No)

TOTALS + = **MOTOR SCORE**
(MAXIMUM) (50) (50) (100)

SENSORY

KEY SENSORY POINTS

	R	L
C2		
C3		
C4		
C5		
C6		
C7		
C8		
T1		
T2		
T3		
T4		
T5		
T6		
T7		
T8		
T9		
T10		
T11		
T12		
L1		
L2		
L3		
L4		
L5		
S1		
S2		
S3		
S4-5		

0 = absent
1 = impaired
2 = normal
NT = not testable

Any anal sensation (Yes/No)

TOTALS + = **PIN PRICK SCORE** (max: 112)
(MAXIMUM) (56) (56) (56) (56)

TOTALS + = **LIGHT TOUCH SCORE** (max: 112)
(MAXIMUM) (56) (56) (56) (56)

NEUROLOGICAL LEVEL

The most caudal segment with normal function

	R	L
SENSORY		
MOTOR		

COMPLETE OR INCOMPLETE?

Incomplete = Any sensory or motor function in S4-S5

ASIA IMPAIRMENT SCALE

ZONE OF PARTIAL PRESERVATION

Caudal extent of partially innervated segments

	R	L
SENSORY		
MOTOR		

评估量表.

Assessment scales

- ① • 身体功能、结构损伤程度评估 (旧称: 残损)
 - Measurement of **impairment of body functions or structure** (previously impairment*)
- ② • 活动受限程度评估 (旧称: 残疾)
 - Measurement of **activity limitation** (previously disability*)
 - Measurement of **participation restriction** (previously handicap*)
- ③ • 社会参与受限评估 (旧称: 残障)
 - Measurement of **quality of life** or health status
- ④ • 生活质量或者健康状况评估

* ICIDH: *International Classification of Impairment, Disability and Handicap*

① 损伤 (残损) 评估量表

Assesment scales for impairment

- 量化疾病对人体解剖结构、功能系统造成的影响
- Quantify the effect of diseases on anatomical structures or functional systems of human organism
- Most common scales:
 - 常用量表:
 - Motricity Index
 - 肌力指数
 - ASIA (measures the level of spinal cord injuries)
 - ASIA量表 (评估脊髓损伤平面)
 - NIH (measures the neurological status of post-stroke patients)
 - Ashworth Scale (measures the level of spasticity of a muscular group)
 - NIH卒中量表 (评估脑卒中后神经状态)
 - Ashworth量表 (评估肌群痉挛程度)

② 活动受限 (残疾) 评估表

Assessment scales for disability

- 评估日常生活能力(ADL)以及工具性日常生活能力(IADL*)
 - Assess the activities of daily living (ADL) or the instrumental activities of daily living (IADL*)
 - The ADL tools measure personal activities like:
 - dressing up
 - eating
 - bathing
 - personal care
 - The IADL tools measure the patient's capabilities of:
 - adapt to the environment
 - house keeping
 - shopping
 - Most common scales: 常用量表:
 - Barthel Index Barthel指数
 - FIM 功能独立性评定FIM
 - PULSES Profile PULSES量表
- ADL评估工具评估个人基础活动, 例如:
 - 穿衣
 - 进食
 - 洗澡、
个人照料
 - IADL工具评估患者下面这些能力:
 - 适应环境
 - 家务
采购

*Instrumental activities of daily living (IADL) are those activities that allow an individual to live independently in a community. Although not necessary for functional living, the ability to perform IADLs can significantly improve the quality of life. The major domains of IADLs include cooking, cleaning, transportation, laundry, and managing finances.

Barthel 指数

Barthel Index (1965)

Barthel指数评估10项常见日常生活活动，包含基础移动能力：

The Barthel Index assesses ten common ADLs, including basic mobility:

- Feeding • 进食
- Bathing • 洗澡
- Grooming • 梳洗，仪容整理
- Dressing • 穿衣
- Bowel • 大便控制
- Bladder • 小便控制
- Toilet use • 使用厕所
- Transfers bed-to-chair-and-back • 床-椅子之间转移往返
- Mobility on level surfaces • 平地移动
- Stair negotiation • 上下楼梯

Barthel Index

这套评估一般几分钟就完成，可以作为康复评估的一部分。

- Barthel指数衡量受试者完成10项移动、自理生活项目，需要多大程度协助
- 根据完成每一项耗时、需要的身体辅助，决定该项得分
- 十个项目分别打分；所有功能项目分数相加得到总分。评估人员通过量化患者表现，评价患者功能障碍。

Typically, the assessment takes only **minutes** to complete and can be a part of a rehabilitation evaluation.

- The Barthel Index measures the **degree of assistance** required by an individual on ten mobility and self-care ADL items
- The time taken and physical assistance required to perform each item are used to determine the assigned **value** of each item
- The ten items are scored with a **number of points**, and then a final score is calculated by summing the points awarded to each functional skill. This allows the examiner to measure a patient's functional disability by quantifying their performance.

Barthel 指数计分

Barthel Index scoring

- 每一项打分依据：能否独立完成、需要帮助、完全依赖他人
- Each item is scored based on whether or not the individual can perform a task or activity independently, with assistance or if they are fully dependent:
 - **0 = unable** • 0 = 无法完成
 - **1 = needs assistance/help** • 1 = 需要协助
 - **2 = independent** • 2 = 独立完成
- The ten items are summed and x 5 to get a total score out of 100. • 10项分数相加乘以5，得到满分100的总分。
- Proposed guidelines for interpreting Barthel scores are as follows: Barthel分数参考解读标准如下：
 - scores of **0-20** indicate “**total**” dependency 0-20分：完全依赖
 - scores of **21-60** indicate “**severe**” dependency 21-60分：重度依赖
 - scores of **61-90** indicate “**moderate**” dependency 61-90分：中度依赖
 - scores of **91-99** indicate “**slight**” dependency 91-99分：轻度依赖

<https://www.sralab.org/sites/default/files/2017-07/barthel.pdf>

THE BARTHEL INDEX

Patient Name: _____
Rater Name: _____
Date: _____

Activity	Score
FEEDING 0 = unable 5 = needs help cutting, spreading butter, etc., or requires modified diet 10 = independent	_____
BATHING 0 = dependent 5 = independent (or in shower)	_____
GROOMING 0 = needs to help with personal care 5 = independent face/hair/teeth/shaving (implements provided)	_____
DRESSING 0 = dependent 5 = needs help but can do about half unaided 10 = independent (including buttons, zips, laces, etc.)	_____
BOWELS 0 = incontinent (or needs to be given enemas) 5 = occasional accident 10 = continent	_____
BLADDER 0 = incontinent, or catheterized and unable to manage alone 5 = occasional accident 10 = continent	_____
TOILET USE 0 = dependent 5 = needs some help, but can do something alone 10 = independent (on and off, dressing, wiping)	_____
TRANSFERS (BED TO CHAIR AND BACK) 0 = unable, no sitting balance 5 = major help (one or two people, physical), can sit 10 = minor help (verbal or physical) 15 = independent	_____
MOBILITY (ON LEVEL SURFACES) 0 = immobile or < 50 yards 5 = wheelchair independent, including corners, > 50 yards 10 = walks with help of one person (verbal or physical) > 50 yards 15 = independent (but may use any aid; for example, stick) > 50 yards	_____
STAIRS 0 = unable 5 = needs help (verbal, physical, carrying aid) 10 = independent	_____
TOTAL (0-100):	_____

Barthel Index scoring

打分可以通过直接观察评估，也可以访谈患者、家属、护理人员获取可靠信息。研究表明评估者的临床经验、常识也可以辅助打分。

- Scores can be assigned either via direct assessment / **observation** or from reliable **interviews** with the patient, family, or staff. Research also suggests that the examiner's "common sense" and clinical experience can be used to assign scores
 - **The higher the score, the more independent the patient is** in completing the measured ADLs. Higher scores also indicate the patient is more likely to return home, with varying degrees of assistance, following hospital discharge
 - **The lower the score, the more dependent the patient is** with ADL completion, and the more skilled care will be required at discharge
 - most studies use a score of 60/61 (moderate dependency) as a cutting point
- 分数越高，患者日常生活越独立。高分代表出院之后，即便需要一定帮助，回家生活的可能性更大。
- 分数越低，患者生活越依赖他人；出院后越需要专业护理。

<https://www.sralab.org/sites/default/files/2017-07/barthel.pdf>

THE BARTHEL INDEX

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Date: _____

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BOWELS 0 = incontinent (or needs to be given enemas) 5 = occasional accident 10 = continent	_____
BLADDER 0 = incontinent, or catheterized and unable to manage alone 5 = occasional accident 10 = continent	_____
TOILET USE 0 = dependent 5 = needs some help, but can do something alone 10 = independent (on and off, dressing, wiping)	_____
TRANSFERS (BED TO CHAIR AND BACK) 0 = unable, no sitting balance 5 = major help (one or two people, physical), can sit 10 = minor help (verbal or physical) 15 = independent	_____
MOBILITY (ON LEVEL SURFACES) 0 = immobile or < 50 yards 5 = wheelchair independent, including corners, > 50 yards 10 = walks with help of one person (verbal or physical) > 50 yards 15 = independent (but may use any aid; for example, stick) > 50 yards	_____
STAIRS 0 = unable 5 = needs help (verbal, physical, carrying aid) 10 = independent	_____
TOTAL (0-100):	_____

⑦ 社会参与(残障)评估量表

Assessment scales for handicap

- Assess psychological, social and economical aspects of the patient
 - 评估患者心理、社会、经济层面情况
- Most common scales:
 - 常用量表:
 - CHART Questionnaire (Craig Handicap Assessment & Reporting Technique)
 - CHART问卷 (Craig残障评估报告量表)
 - CIQ (Community Integration Questionnaire)
 - CIQ社区融合问卷

CHART Questionnaire (Craig Handicap Assessment & Reporting Technique)

- Designed to assess how people with disabilities function as active members of their communities • 用来评估残障人士作为社区一份子的社会活动能力
- 6 domains: 6个维度:
 - Physical independence • 身体独立能力
 - Cognitive independence • 认知独立能力
 - Mobility • 移动能力
 - Occupation • 工作职业
 - Social integration • 社会融合
 - Economic self-sufficiency • 经济自给自足
- 32 items in total 一共32个条目
- Scores on each subscale (domain) range from 0-100 with total CHART score ranging from 0-600. 每个子维度得分0-100; CHART总分区间0-600。
- **Higher scores indicate a lesser degree of handicap or greater degree of social and community participation.** 分数越高, 残障程度越低, 社区社会参与度越高。

打分示例

Ex. PHYSICAL INDEPENDENCE scoring with CHART

身体独立维度，得100分代表完全没有残障，可以维持常规独立生活。原本自己可以做的事现在需要经常性间断的协助，代表存在一定残障。

Revised Craig Handicap Assessment and Reporting Technique Scoring Form

		PHYSICAL INDEPENDENCE
1.	How many hours in a typical 24-hour day do you have someone with you to provide physical assistance for personal care activities such as eating, bathing, dressing, toileting and mobility? _____ hours paid assistance _____ hours unpaid (family, others)	Total the hours of paid and unpaid care. + _____
2.	Not including any regular care as reported above, how many hours in a <u>typical month</u> do you occasionally have assistance with such things as grocery shopping, laundry, housekeeping, or infrequent medical needs because of the disability? _____ hours per month	Divide the hours of occasional care by 30. $(\text{_____} / 30)$ = _____
		Add the sums of "A" and "B".
3.	Who takes responsibility for instructing and directing your attendants and/or caregivers? _____ Self _____ Someone Else _____ Not applicable, does not use attendant care	If the respondent instructs and directs his/her own attendants or caregivers, multiply the answer of "C" by 3. X 3 or 4 = _____
		If someone other than the respondent instructs and directs the attendants or care givers, multiply the answer of "C" by 4. _____
		100 minus _____
	Subtract the total in "D" from 100.	sum from "D" above = _____

PHYSICAL INDEPENDENCE SCORE

In the **Physical Independence** dimension, a score of 100 indicates no handicap in an individual's ability to sustain a customarily effective independent existence. The need for regular or periodic assistance for activities which used to be performed independently is indicative of some degree of handicap.

SCORE COMPUTATION:

- Add the hours of paid to unpaid care in Question #1.
- Divide the hours of occasional care in Question #2 by 30. This computation results in an equivalent of hours per day of occasional care.
- Add the sums of "A" and "B".
- If the respondent instructs and directs his/her own attendants or caregivers (Question #3), multiply the answer of "C" by 3. If someone other than the respondent instructs and directs the attendants or care givers, multiply the answer of "C" by 4.
- Subtract the total in "D" from 100. The remainder will be the **Physical Independence** dimension score.

④ 生活质量评估量表

Assessment scales for the quality of life

- 站在患者角度评估疾病带来的影响；从心理、生理层面测量患者自我主观感受相关指标
- Assess the impact of disease from the patient's point of view and measure the parameters related to the perception that the patient has of him/herself, from the psychological and physical point of view
- Most common scales:
 - 常用量表：
 - SIP (Sickness Impact Profile) • SIP疾病影响概况量表
 - SF-36 (36-Item Short-Form) • SF-36健康调查简表（36条目）
 - MRFA (Medical Rehabilitation Follow Along) • MRFA医疗康复随访量表

评估量表的特征

Characteristics of assessment scales

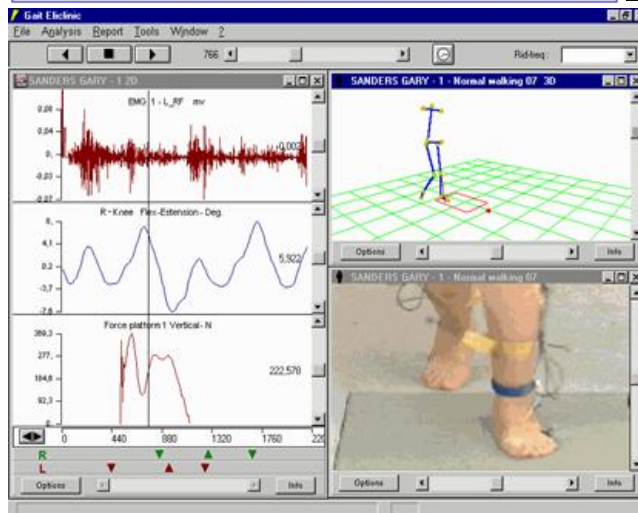
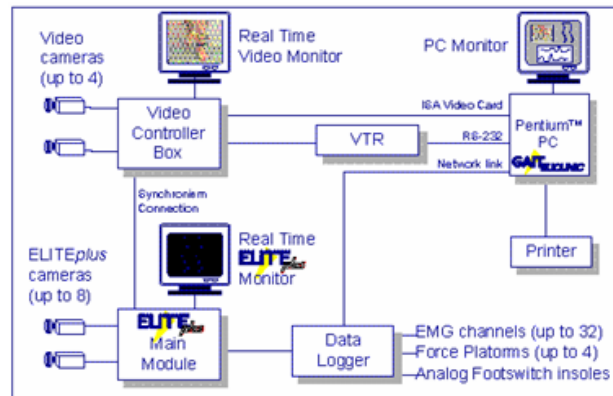
可信度：测量结果不受误差干扰的程度

可重复性：多次测量、多名评估者在相近条件操作，结果可以复现的程度

- **Reliability**: how much the measure is free from errors
 - Repeatability**: how much the measure can be reproduced, when executed more times or by more examiners, appropriately and in similar conditions
- **Inter-examiner reproducibility** : level of overlapping between scores obtained by different trained examiners
- **Test-retest reproducibility**: level of variability of scores obtained in repeated measurements in a so-called stable status
 - 评估者间可重复性：多名受过训练评估人得到分数的一致程度
 - 重测可重复性：受试者状态稳定时，重复测量得到分数的波动大小

运动分析：对人体运动开展定量研究

Motion analysis: quantitative study of human movement



- 运动学数据：动作捕捉系统测得的人体运动信息
- 动力学数据：测力台与各类传感器测得力、力矩
- 肌电（EMG）数据：用电极采集，反映肌肉激活状态的电信号
- **Kinematic data:** human movements, measured with motion capture systems;
- **Dynamic data:** forces and angular moments, measured with force platforms and other sensors;
- **Electro-myographic (EMG) data:** electrical signals of muscle activations, measured with electrodes

Summary

Questions ?

- Learnt some terminology about human anatomy
 - Anatomy and role of vertebral column
 - Spinal cord injuries and related paralyses
 - Scales for assessment of
 - Impairment
 - Disability
 - Handicap
 - Quality of life
- 学习人体解剖相关术语
 - 脊柱的解剖结构与功能
 - 脊髓损伤以及对应的各类瘫痪
- 评估各类量表：
- 残损（身体结构功能损伤）
 - 残疾（活动受限）
 - 残障（社会参与受限）
 - 生活质量